ChatGPT 5.2

Tutorial_2.pdf
PDF

Task: Generate a lecture scribe.

Purpose- A lecture scribe is a faithful, exam-ready reconstruction of what was taught in class. The scribe should function as reference material suitable for preparing for a reading-based / closed-notes exam.

The context is fixed and authoritative.
Use ONLY the provided Tutorial-2 document.
Everything included must be traceable to the provided context.
Do not use external knowledge or introduce material not present in the context.

Prompting Style: PROMPTING STYLE 3: CHAIN-OF-THOUGHT (CoT) PROMPTING.

Chain-of-Thought Requirement- While writing the scribe, reason step by step. Reasoning may appear in the final scribe to improve clarity for exam preparation. All reasoning must come only from the provided context.

Scope Clarification for This Tutorial-
This scribe must NOT be written question-wise.
Do NOT restate the questions.
Do NOT provide direct answers.
Do NOT compute numerical values.
Do NOT present final probabilities or simplified results.

Instead, extract and organize the underlying mathematical content required to solve the tutorial problems.

Accordingly, include Definitions and notation that are necessary. Assumptions and conditions underlying each distribution. Statements of relevant theorems or results. General derivations and identities required (without substituting numerical values). All probability distributions that appear (Uniform, Normal, Exponential, Poisson, etc.). General methods implied by the problems (conditional probability, independence, memoryless property, expectation minimization, transformation of random variables, order statistics, joint distributions, etc). Clear logical flow between ideas.

The scribe should allow a student to-
- Recall definitions and notation
- Reconstruct derivations independently
- Follow the mathematical reasoning step by step
- Apply the formulas during revision to similar problems

Do NOT include:
- A solution manual
- Worked answers to specific questions
- New examples
- Added intuition not present in context
- Material not discussed in the Tutorial-2 document

Explanation Constraints-
Explain definitions and assumptions explicitly.
Provide step-by-step derivations only at the general formula level.
Do not provide alternative reasoning paths.

Structural Requirements- Organize the scribe by concepts rather than by question number. Maintain formal mathematical structure suitable for exam revision.

Tone- Maintain a formal, academic tone suitable for exam preparation. Prioritize correctness and faithfulness to the context.

Finally:

Write the lecture scribe such that a student studying only this document can reliably revise the concepts and methods required to solve the tutorial problems independently.

Lecture Scribe

CSE 400: Fundamentals of Probability in Computing

Tutorial–2

Source: Tutorial-2 Document Tutorial_2

This scribe organizes and reconstructs the mathematical concepts, definitions, distributions, and general derivation techniques required to solve the problems in Tutorial–2. It is structured conceptually rather than question-wise and contains only material traceable to the provided document.

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1. Continuous Uniform Distribution on an Interval

1.1 Definition

A continuous random variable X is **Uniform on (a,b)** if:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

The cumulative distribution function (CDF):

$$F_X(x) = \begin{cases} 0, & x \le a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x \ge b \end{cases}$$

The expectation:

$$E[X] = \frac{a+b}{2}$$

1.2 Random Point on a Line Segment

If a point is chosen at random on a line segment of length L, this means:

$$X \sim \text{Uniform}(0, L)$$

A geometric probability involving ratios (e.g., shorter/longer segment) requires:

- Expressing the ratio in terms of X
- Identifying regions of the sample space where the inequality holds
- Computing probability via integration over the appropriate interval

For continuous variables:

$$P(A) = \int_{\text{region}} f_X(x)\, dx$$

Since the density is constant for uniform distributions, probability reduces to:

$$\frac{\text{length of favorable region}}{\text{total length}}$$

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2. Normal (Gaussian) Distribution and Bayesian Classification

2.1 Definition

A random variable $X \sim N(\mu, \sigma^2)$ has pdf:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Key parameters:

- Mean: μ
- Variance: σ^2

2.2 Mixture Models and Prior Probabilities

When a point is drawn from two possible populations:

- With probability α, it comes from one distribution
- With probability $1-\alpha$, from another

This forms a **mixture distribution**:

$$f(x) = \alpha f_1(x) + (1-\alpha) f_2(x)$$

2.3 Posterior Probability (Bayes' Rule)

If A and B partition the sample space:

$$P(A \mid x) = \frac{f_A(x)P(A)}{f_A(x)P(A) + f_B(x)P(B)}$$

Equal error probability condition implies balancing posterior probabilities.

Thus, one sets:

$$P(A \mid x) = P(B \mid x)$$

leading to an equation in prior probability α.

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3. Minimization of Expected Absolute Deviation

We are interested in minimizing:

$$E[|X - a|]$$

3.1 General Form

$$E[|X - a|] = \int_{-\infty}^{\infty} |x - a| f_X(x)\, dx$$

Split at $x = a$:

$$= \int_{-\infty}^{a} (a - x) f_X(x)\, dx + \int_{a}^{\infty} (x - a) f_X(x)\, dx$$

Differentiating with respect to a gives:

$$\frac{d}{da} E[|X - a|] = P(X \le a) - P(X \ge a)$$

Setting derivative to zero:

$$P(X \le a) = \frac{1}{2}$$

Thus:

The minimizer of $E[|X - a|]$ is the median of X.

3.2 Uniform Case

For $X \sim \text{Uniform}(0, A)$:

Median = $\frac{A}{2}$

3.3 Exponential Case

If $X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

The median m satisfies:

$$P(X \leq m) = \frac{1}{2}$$

$$1 - e^{-\lambda m} = \frac{1}{2}$$

This determines the optimal location.

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4. Exponential Distribution and Memoryless Property

4.1 Definition

If $X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$F_X(x) = 1 - e^{-\lambda x}$$

4.2 Memoryless Property

For exponential random variables:

$$P(X > t + s \mid X > t) = P(X > s)$$

Proof:

$$\frac{P(X > t + s)}{P(X > t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda s}$$

4.3 Mixture of Exponentials

If:

- With probability p_i, rate is λ_i

Then survival probability:

$$P(X > t) = \sum_i p_i e^{-\lambda_i t}$$

Conditional survival:

$$P(X > t + s \mid X > t) = \frac{\sum_i p_i e^{-\lambda_i (t+s)}}{\sum_i p_i e^{-\lambda_i t}}$$

This is **not memoryless** unless rates are identical.

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5. Poisson Distribution and Conditional Structure

5.1 Definition

$X \sim \text{Poisson}(\lambda)$

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Mean and variance:

$$E[X] = \lambda$$

5.2 Sum of Independent Poissons

If:

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

and independent, then:

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

5.3 Conditional Distribution Given Total

$$P(X=k \mid X+Y=n) = \frac{P(X=k, Y=n-k)}{P(X+Y=n)}$$

Using independence:

$$= \frac{\frac{e^{-\lambda_1}\lambda_1^k}{k!}\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$$

This simplifies to a Binomial-type structure.

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6. Transformation of Random Variables

6.1 CDF Method

If $Y = g(X)$:

$$F_Y(y) = P(g(X) \le y)$$

Then differentiate:

$$f_Y(y) = \frac{d}{dy}F_Y(y)$$

6.2 Piecewise Transformations

If g is not one-to-one:

$$f_Y(y) = \sum_{x:g(x)=y} \frac{f_X(x)}{|g'(x)|}$$

Valid when inverse branches exist.

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7. Binomial Approximation via Independent Movements

If each period:

- Probability p of upward movement
- Independent across periods

Then after n periods:

Number of upward movements:

$$X \sim \text{Binomial}(n, p)$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Large n approximation:

$$X \approx N(np, np(1-p))$$

Probability involving percentage increase translates into inequality in X.

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8. Conditional Probability with Exponential Variables

Using survival function:

$$P(X > t) = e^{-\lambda t}$$

Conditional probability:

$$P(X \geq t + s \mid X \geq t) = \frac{P(X \geq t + s)}{P(X \geq t)}$$

Apply exponential survival form.

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9. Joint Transformation to Radial Variable

Let X and Y independent with pdfs $f_X(x), f_Y(y)$.

Define:

$$Z = \sqrt{X^2 + Y^2}$$

To obtain pdf:

1. Use joint density:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

2. Transform to polar coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta$$

Jacobian:

$$J = r$$

3. Integrate over angle:

$$f_Z(r) = \int_0^{2\pi} f_X(r\cos\theta) f_Y(r\sin\theta)\, r\, d\theta$$

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10. Order Statistics and Joint Distributions

Let:

$$X_1, X_2, \ldots$$

be i.i.d. with CDF F.

Define:

$$M_n = \max(X_1, \ldots, X_n)$$

10.1 Distribution of Maximum

$$P(M_n \leq x) = P(X_1 \leq x, \ldots, X_n \leq x)$$

Using independence:

$$= [F(x)]^n$$

Thus:

$$F_{M_n}(x) = [F(x)]^n$$

10.2 Joint Distribution of M_n and M_{n+1}

Observe:

$$M_{n+1} = \max(M_n, X_{n+1})$$

Joint event:

$$P(M_n \leq x, M_{n+1} \leq y)$$

Case distinction:

- If $y < x$: probability is zero
- If $y \geq x$:

$$P(M_n \le x, X_{n+1} \le y) = [F(x)]^n F(y)$$

This characterizes the joint CDF structure.

Summary of Core Concepts

This tutorial requires mastery of:

- Uniform distribution and geometric probability
- Normal distributions and mixture models
- Bayes' rule for posterior probabilities
- Minimization of expected absolute deviation (median property)
- Exponential distribution and memorylessness
- Mixture of exponentials
- Poisson distribution and conditional structure
- Binomial modeling of repeated independent trials
- Normal approximation to Binomial
- Transformation of random variables (CDF and Jacobian methods)
- Order statistics and maxima distributions

This document contains all definitions, distributions, identities, and derivation frameworks necessary to independently reconstruct solutions to the Tutorial–2 problems.

Sources